AQ2.2: Activity Questions 2- Not Graded

This assignment will not be graded and is only for practice.

Level 1:

1 point

Consider a system of linear equations as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A ($i.e.$, $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b, for $i = 1, 2, 3$. If $det(A_{x_i}) = 0$ for $i = 1, 2, 3$, then which of the following is (are) true?

- [ ] $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ must be a solution.
- [ ] $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a solution if $det(A) \neq 0$.
- [ ] If $det(A) = 0$, then we cannot conclude about the solution using Cramer's rule.
- [ ] If $det(A) \neq 0$, then $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$x = $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a solution if $det(A) \neq 0$det(A) = 0.

If $det(A) = 0$det(A) = 0, then we cannot conclude about the solution using Cramer's rule.

If $det(A) \neq 0$det(A) = 0, then $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$b = $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Consider a system of linear equations

$$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$

x1 + x3 = 1−x1 + x2 − x3 = 1−x2 + x3 = 1

Let matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$b = $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Let A_{x_i} Axi be the matrix obtained by replacing the ii-th column of AA (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by bb, for $i = 1, 2, 3$i = 1, 2, 3.

(Use the above information to answer questions 2 and 3)

1 point

Choose the set of correct options

- [ ] $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$Ax1 = $\begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

- [ ] $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$Ax1 = $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

- [ ] $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$Ax2 = $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

A_{x_3} = [1 0 1 −1 1 1 0 −1 1]Ax3 = [1−1001−1111]

No, the answer is incorrect.
Score: 0
Accepted Answers:

A_{x_1} = [1 0 1 1 1 −1 1−1 1]Ax1 = [11101−11−11]

A_{x_3} = [1 0 1 −1 1 1 0 −1 1]Ax3 = [1−1001−1111]

1 point
Choose the set of correct options.

- [ ] x_1 = − 2x1 = −2.
- [ ] x_2 = − 2x2 = −2.
- [ ] x_3 = 3x3 = 3.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
x_1 = − 2x1 = −2.
x_3 = 3x3 = 3.

Level 2:

Consider the system of linear equations $Ax = b$Ax = b, where A = [1 a 0 a 1 2 1 0 a]A = [1a1a1002a], x = [x_1 0 x_3]

x = [x10x3], and b = [1 0 0]b = [100], and the solution for xx is partially known. What is the value of

a^2 a2, if $a \neq 0, \sqrt{3}, -\sqrt{3}$a = 0, 3 $\sqrt{}$, − 3 $\sqrt{}$

is given? [Hint: Observe that the second row of the vector xx is given as 00, which implies that x_2 x2 is known, where x = [x_1 x_2 x_3]x = [x1x2x3]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider the system of linear equations Ax = bAx = b, where A = [1 $2a$ $3a$ + 2 1 + 3]A = [13a + 22a + 31], x = [x_1 0]x = [x10], and b = [1 1]b = [11], and the solution for xx is partially known.

What is the value of 3a3a, if $det(A) \neq 0$det(A) = 0 is given?

No, the answer is incorrect.
Score: 0